

Brainstorming the solar radiation story

Divergent ideas, evidence questions, and early chart sketches before implementation.

Central audience question

How can Hong Kong planners understand when solar radiation is strongest, what weather conditions explain it, and when extreme opportunities occur?

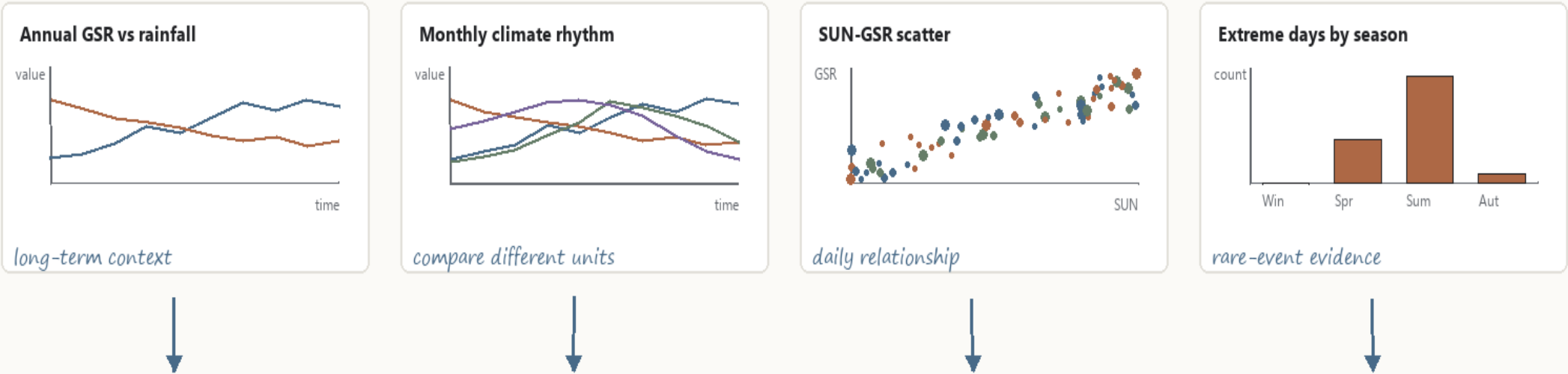
Working hypotheses

- H1 GSR and rainfall move in opposite directions.
- H2 Sunshine duration is the strongest positive association; humidity is negative.
- H3 Extreme GSR days cluster in high-sunshine, low-rainfall seasons.

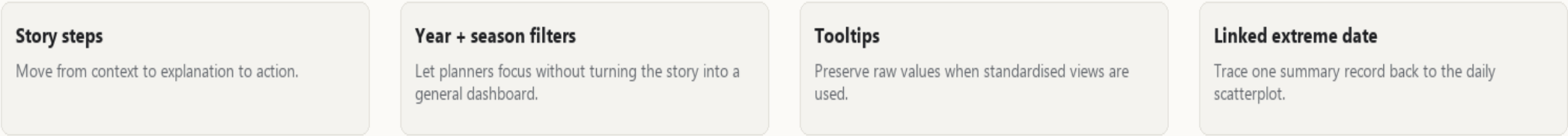
Ideas removed or simplified

- ~~Map: one station cannot support a spatial claim~~
- ~~Raw daily line chart: too noisy across 11,007 days~~
- ~~Dense pairplot: too analytical for the target audience~~

Candidate visual encodings



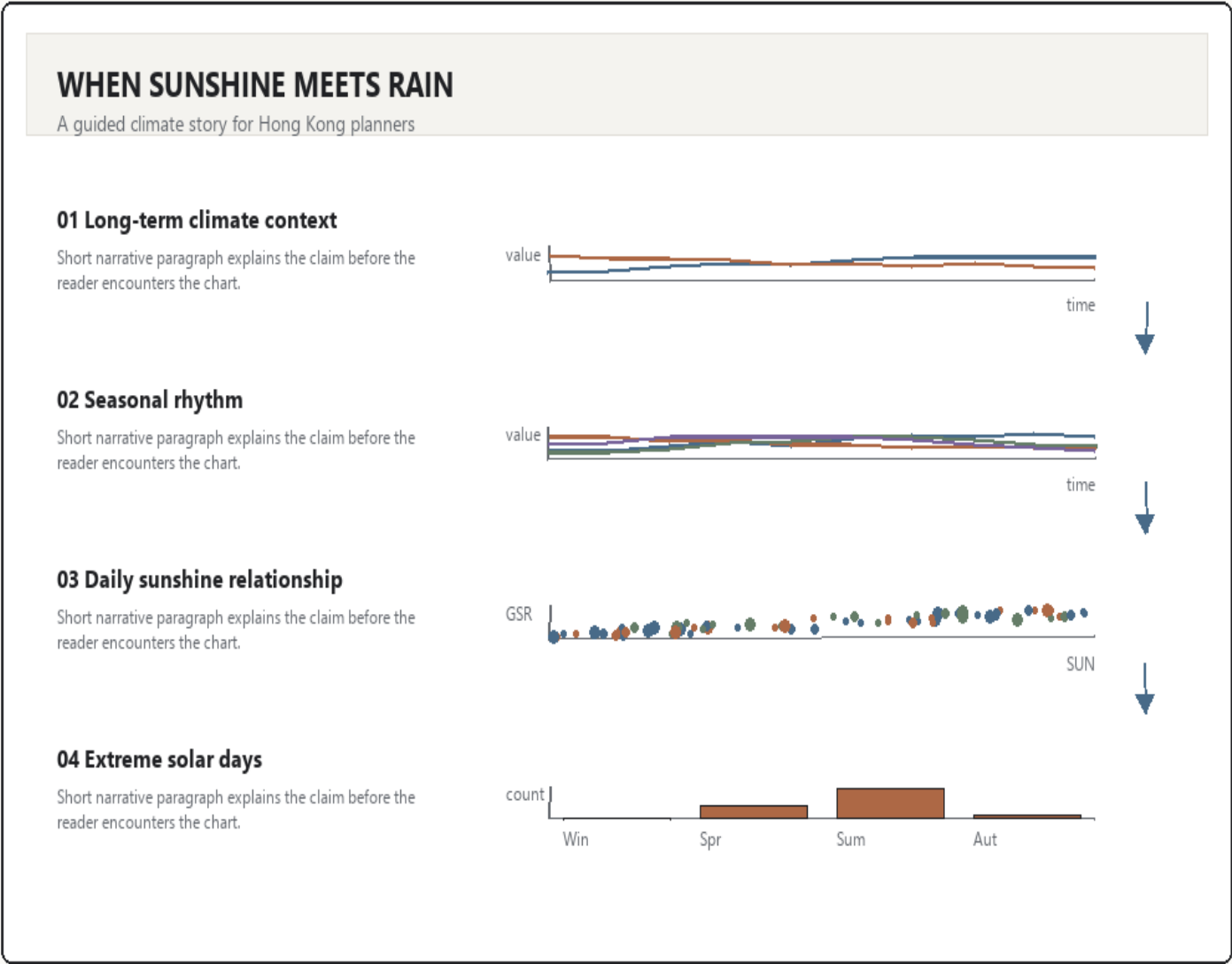
Possible interactions and narrative structure



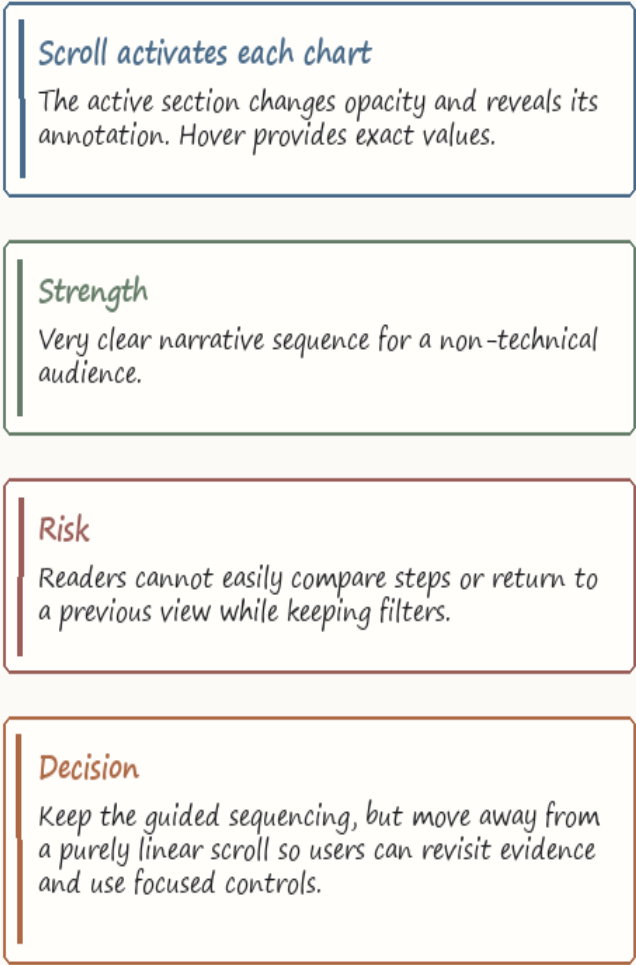
Alternative A: Scrollytelling climate story

A linear article-style experience where each scroll section introduces one visual claim.

Complete interface sketch



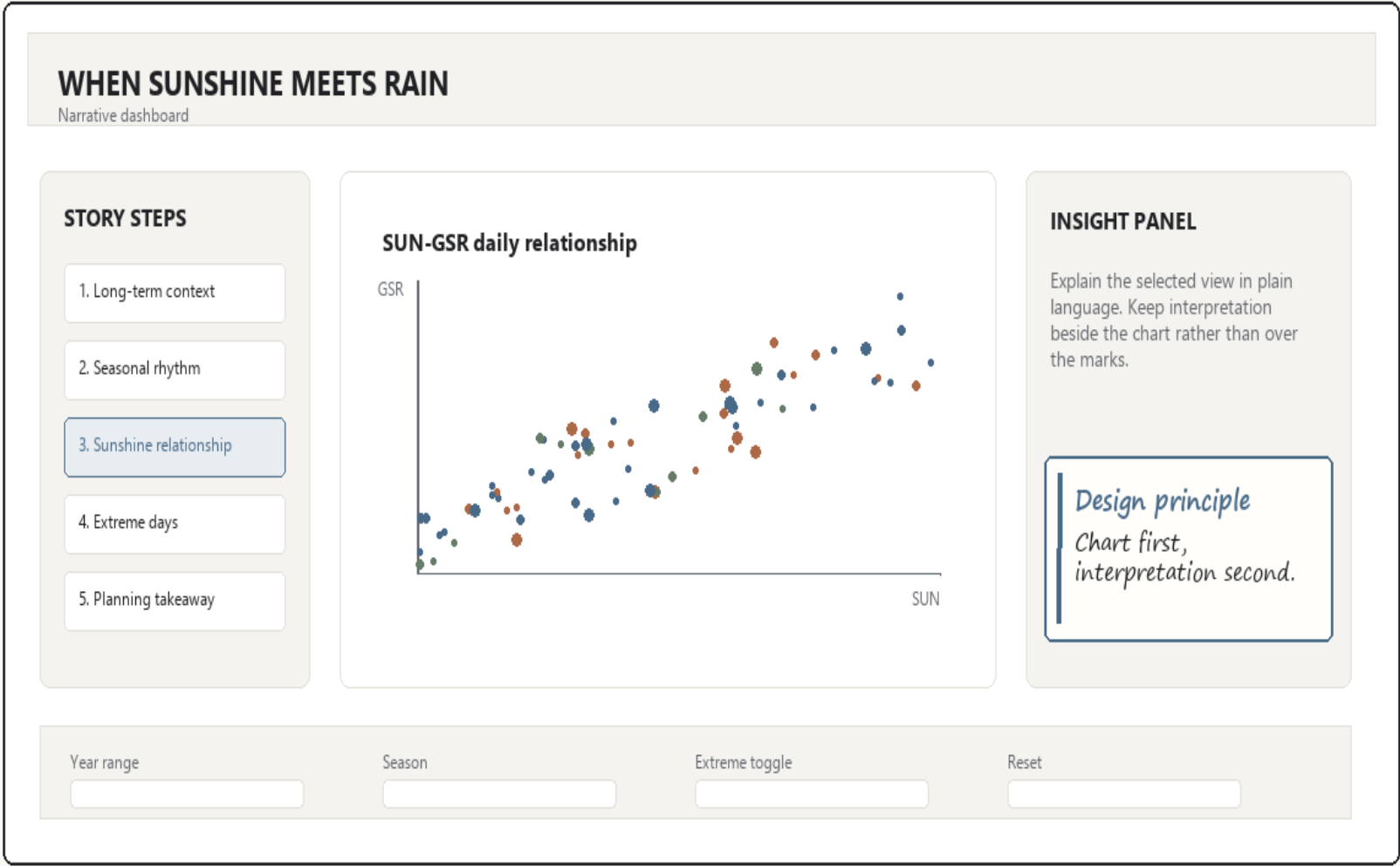
Interaction sketch



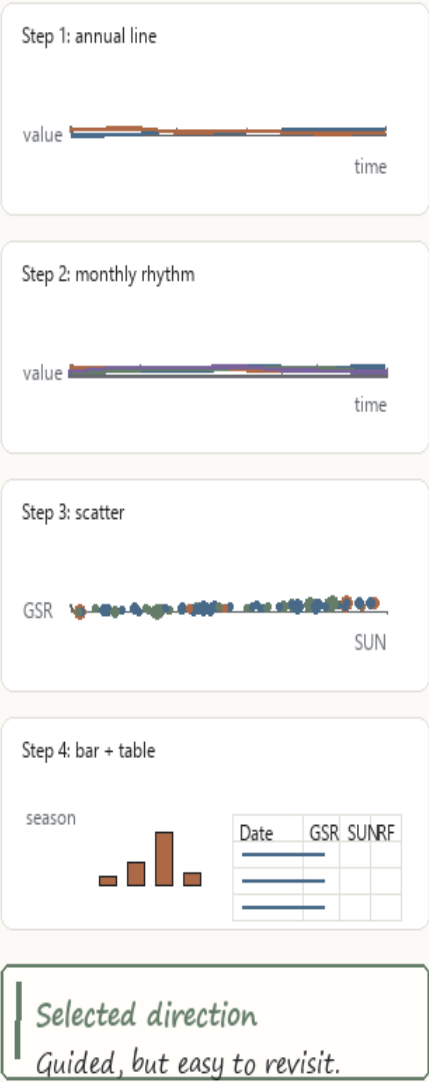
Alternative B: Guided story dashboard

A stable workspace with explicit story steps, a main chart, an insight panel, and focused controls.

Complete interface sketch



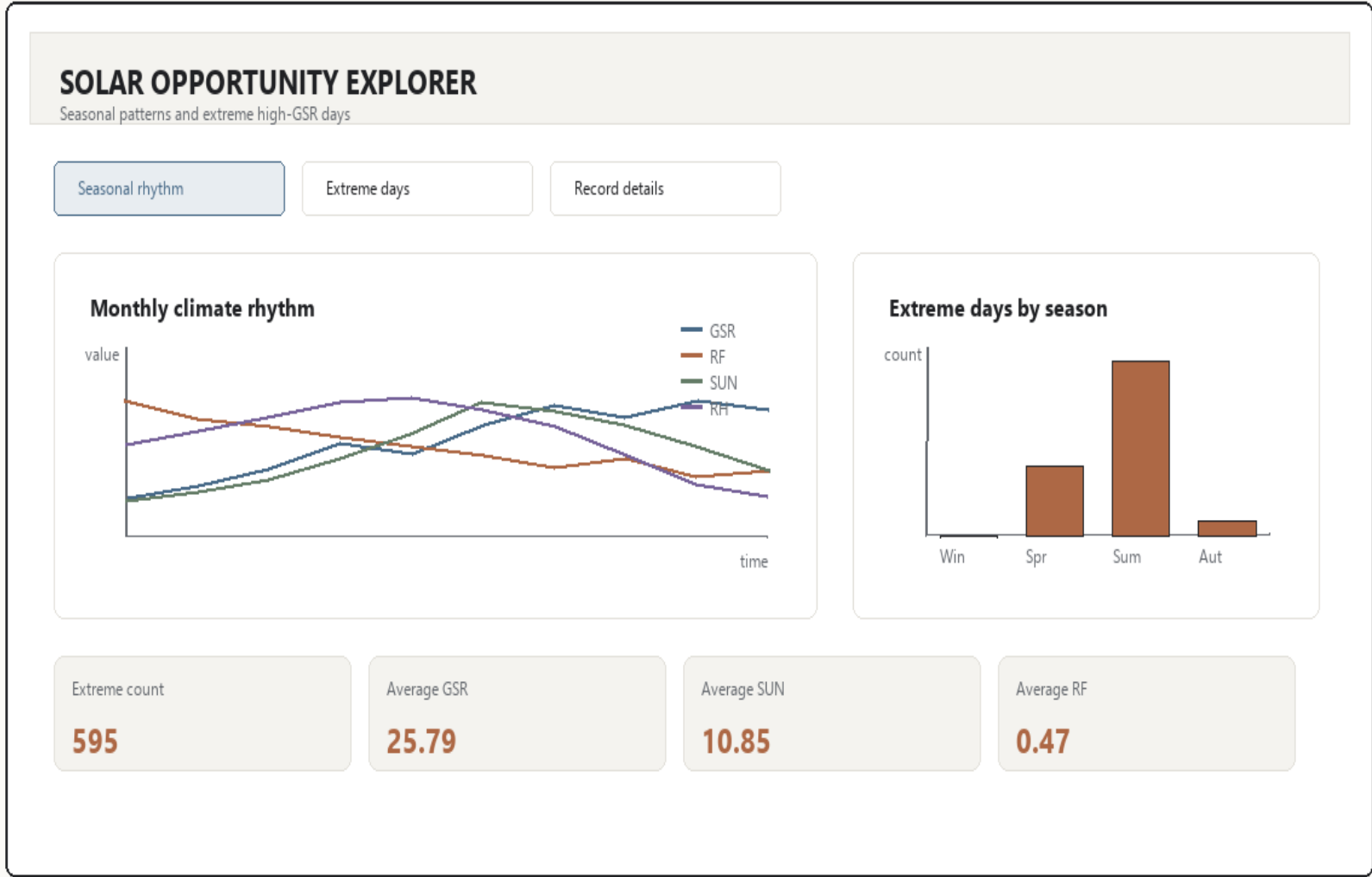
Step-specific chart sketches



Alternative C: Seasonal and extreme-event explorer

A task-focused concept that prioritises monthly climate rhythm and rare high-radiation days.

Complete interface sketch



Final realisation: Guided five-step narrative

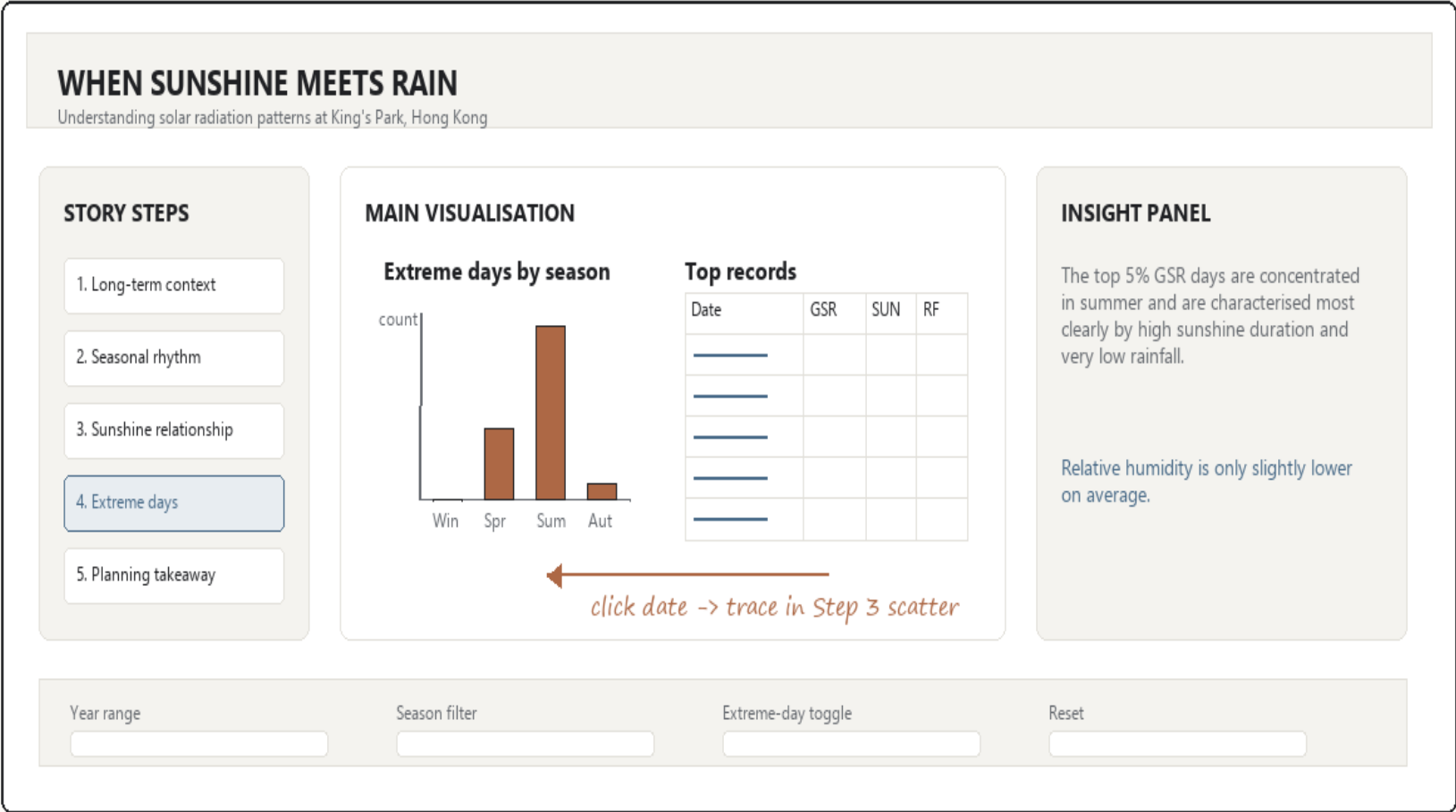
The implementable design combines Sheet 3's stable narrative workspace with Sheet 4's seasonal and extreme-event evidence.

Refinement after Part 1 feedback

Draft chart sketches, interaction arrows, and implementation annotations were added so the design communicates what each story step will actually show.

Evidence-aware refinement

The final wording does not overclaim humidity: extreme days are most clearly high-sunshine, low-rainfall summer events, while humidity is supporting context.



Five-step evidence plan

- 1 Annual line
Test H1 across years
- 2 Monthly rhythm
Show seasonal overlap
- 3 SUN-GSR scatter
Test H2 in daily data
- 4 Bar + table
Test H3 and trace records
- 5 Takeaway cards
Translate evidence for planners

Implementation
D3.js + local CSV + state.